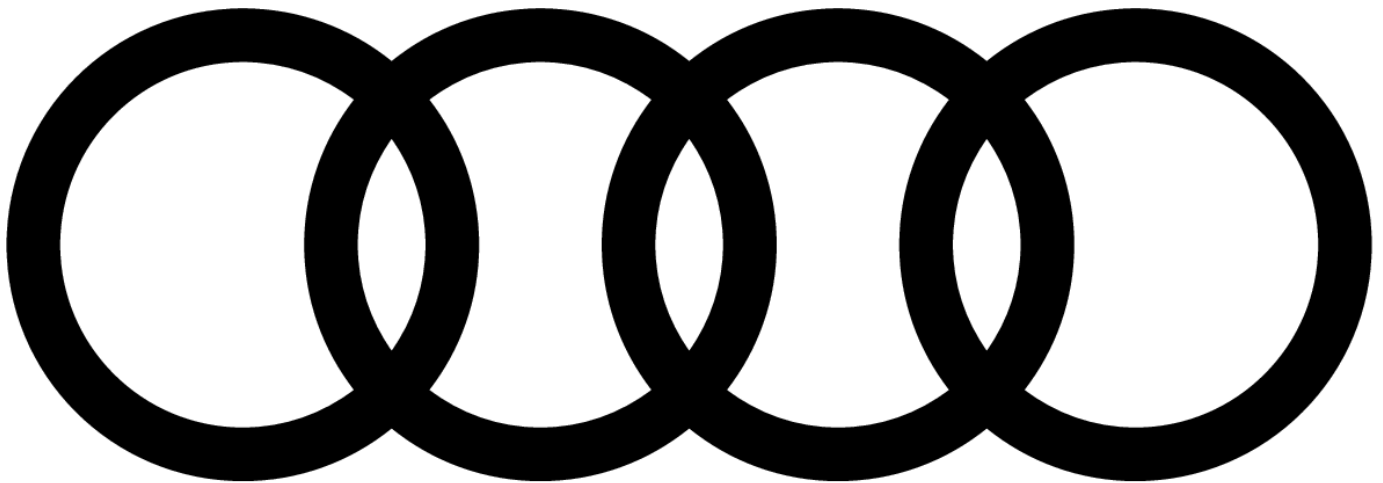


2024



Audi

AUDI Saved More than 60% on Compute Costs and Improved Scalability for Its Car-Configurator Backend Using Karpenter

Learn how AUDI improved its applicatio

[Overview](#) | [Opportunity](#) | [Solution](#) | [O](#)



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63%



2

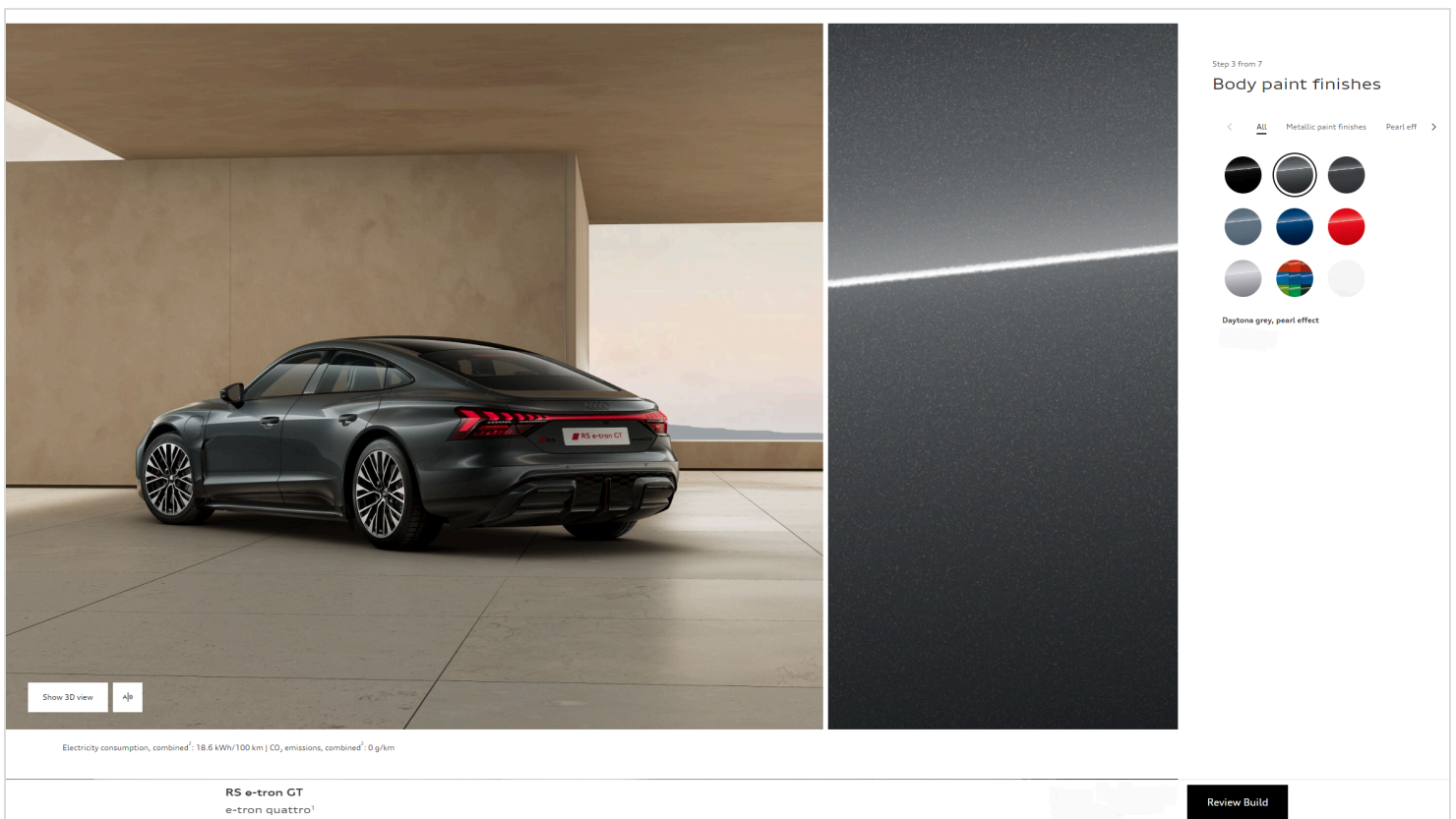
in cost savings

Up to 20%

shorter application startup times using Graviton-based instances compared with on premises

Overview

Global premium automotive manufacturer [AUDI AG](#) (AUDI) helps customers and dealers customize the perfect configurations of its vehicles with its online car-configurator tool. This essential part of the car-buying process is available in over 30 markets worldwide, but AUDI's monolithic, on-premises infrastructure made it inefficient and complex to maintain. By migrating to Amazon Web Services (AWS), AUDI improved the scalability, performance, and cost of its tool.



Opportunity | Improving Scalability, Availability, Efficiency, and Latency Using AWS for AUDI

AUDI, a subsidiary of the Volkswagen Group, manufactures premium automobiles for more than 100 markets worldwide. Using its car configurator, dealers can customize and place orders for their customers. Customers also use the tool to shop and compare vehicles on AUDI's website. The car configurator is the main entry point for customers and is a highly business-critical application.

AUDI's car-configurator backend ran on premises as a monolithic application. But AUDI wanted to make it more scalable and resource efficient to better handle large traffic spikes without overprovisioning resources and to release software updates to individual markets. So, it needed great efficiency, high availability, and scalability. The company was overprovisioned in the on-premises environment and needed to



servers on standby because it would take days or weeks of internal processes to set up completely new environments.

In 2018, AUDI tested its tool in the cloud as a prototype in the North American market. After verifying the benefits of the cloud through the prototype, AUDI migrated its production application to [Amazon Elastic Compute Cloud](#) (Amazon EC2), which provides secure and resizable compute capacity for virtually any workload. The company began deploying its infrastructure using [Terraform](#), an infrastructure automation tool, and [AWS CloudFormation](#), which speeds up cloud provisioning with infrastructure as code. In the cloud, AUDI no longer has to manage equipment and data centers; instead, it can provision resources as needed. The company also addressed the latency issues that its overseas customers experienced by deploying workloads close to these customers using [AWS Regions](#)—physical locations around the world where AWS clusters data centers.

On AWS, development speed and stability have increased, along with developer productivity. “We’re now doing everything with infrastructure as code,” says Florian Königer, PO and system architect at AUDI. The complete environment is built using Terraform, and AWS CloudFormation for some components. AUDI is also simplifying maintenance and empowering developers with more options to set permissions at a granular level and avoid unnecessary overhead. And by applying [blue/green deployments](#), AUDI can stabilize its environment because it can cover resource spikes in the cloud.

To increase agility, deployment speed, and scalability for its application, AUDI migrated to a containerized architecture. This also made it simple to release updates to individual markets. AUDI chose to split its monolithic application into microservices by deploying containers using [Kubernetes](#)—an open-source system for managing containers—and its Cluster Autoscaler feature. The company wanted to further optimize its operational management and increase flexibility while taking advantage of the various Amazon EC2 purchasing options and multiarchitecture solution. So, it decided to migrate to [Karpenter](#), an open-source, high-performance Kubernetes cluster automatic scaler, which was originally built and contributed to the Cloud Native Computing Foundation by AWS.

Solution | Increasing Performance and Flexibility Using Graviton-Based Instances and Karpenter

Using Karpenter, the company can use different Amazon EC2 purchase options, instance types, and sizes, in addition to a mix of instances with different processors, through a single configuration. Thus, AUDI can choose the perfect instance according to the price-capacity-optimized allocation strategy of that inst



purchasing options include [Amazon EC2 On-Demand Instances](#), [Amazon EC2 Spot Instances](#), and [Amazon EC2 Reserved Instances](#).

In 2021, AUDI began using Amazon EC2 instances that are powered by [AWS Graviton processors](#), which are custom-designed server processors developed by AWS to provide up to 40 percent better price performance for cloud workloads running on Amazon EC2. As it expanded its multiarchitecture solution in 2024, the company used the management capabilities of Karpenter to facilitate provisioning AWS Graviton-based Amazon EC2 instances.

AUDI's development team uses a combination of Graviton-based instances, Reserved Instances, and Spot Instances for production workloads. Thus, the company has optimized the price performance of its workloads and reduced compute costs for the car configurator. AUDI uses Karpenter to run about a thousand Kubernetes pods in production. Now, AUDI has each instance directly managed by Karpenter, so it can be even more flexible. By doing provisioning on an instance level instead of an automatic-scaling group level, the company has reduced costs. It saved 63 percent on compute costs and accelerated its application's startup time by up to 20 percent using Graviton-based instances compared with the on-premises system.

AUDI optimized resource consumption as well. By using Karpenter, the company has increased efficiency, especially on weekend schedules to reduce provisioning capacity in development environments.

Outcome | Continuing the Cloud Migration to Improve the Customer Experience

AUDI accelerated its application's startup time, improved scalability, and reduced latency for customers by migrating its car-configurator application to a mix of Amazon EC2 instances and Karpenter. "On AWS, availability has increased compared with our on-premises setup," says Königer. "We also have lower latency and shorter response times."

By migrating to the cloud, the company has optimized its compute infrastructure and improved the experience for both customers and dealers.

About AUDI AG

AUDI AG, a subsidiary of the Volkswagen Group, is a global premium automotive manufacturer that produces premium vehicles for more than 100 markets around the world.

AWS Services Used

AWS Graviton Processor

AWS Graviton is a family of processors designed to deliver the best price performance for workloads running in Amazon Elastic Compute Cloud (Amazon EC2).

[Learn more »](#)



Amazon EC2 Spot Instances

Amazon EC2 Spot Instances let you take advantage of unused EC2 capacity in the AWS cloud and are available at up to a 90% discount compared to On-Demand prices.

[Learn more »](#)

AWS Regions

AWS has the concept of a Region, which is a physical location around the world where we cluster data centers. We call each group of logical data centers an Availability Zone. Each AWS Region consists of a minimum of three, isolated, and physically separate AZs within a geographic area.

[Learn more »](#)

AWS CloudFormation

AWS CloudFormation is a service that helps you model and set up your AWS resources so that you can spend less time managing those resources and more time focusing on your applications that run in AWS.

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EUROPE, MIDDLE EAST, & AFRICA



Continental on AWS

From one of the most recognized tire brands in the world to a connected



scenarios that can be directed to specific locations and environments.

2024

mobility pioneer, Continental has seen success from 1871 to today thanks to its constant evolution. Using Amazon Web Services (AWS), the company has embraced the cloud and accelerated a software-driven vision. It built modular solution Continental Automotive Edge (CAEdge), which connects vehicles to the cloud and provides a virtual



2024

ITALY



Ferrari on AWS

Luxury Italian auto manufacturer Ferrari S.p.A. (Ferrari) has forged a globally recognized legacy that's rooted in tradition and innovation over many decades. To continue to deliver the best possible experiences to customers and dealers, Ferrari has tapped into the power of generative artificial intelligence (AI) by turning to Amazon Web Services (AWS).

2024

EUROPE, MIDDLE EAST, & AFRICA



Audi

AUDI Saved More than 60% on Compute Costs and...

Learn how AUDI improved its application's reliability, scalability, and cost efficiency using Karpenter.

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